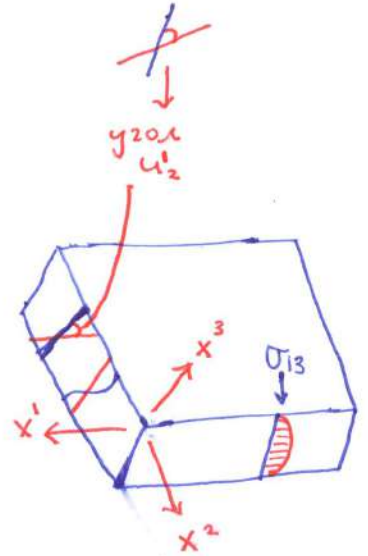
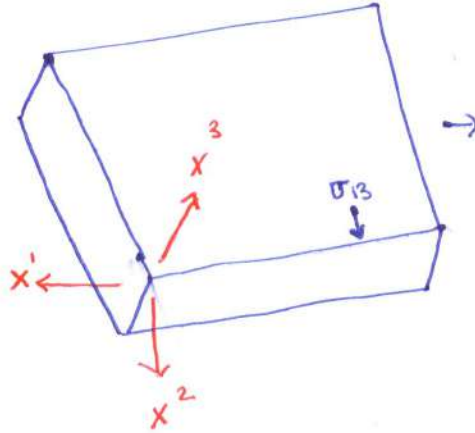
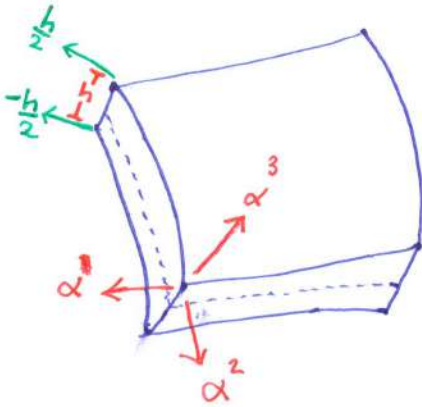


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$$\sigma_{ij} n_j = F_i$$

$$\sigma_{13} = \sigma_{23} = 0$$

$$\Rightarrow \varepsilon_{13} = \varepsilon_{23} = 0$$

$$\begin{aligned} u_1 &= u_1^0(x^1, x^2) + u_1^1(x^1, x^2) x^3 + \frac{1}{2} u_1^2(x^1, x^2) (x^3)^2 + \dots \\ u_2 &= u_2^0(x^1, x^2) + u_2^1(x^1, x^2) x^3 + \frac{1}{2} u_2^2(x^1, x^2) (x^3)^2 + \dots \\ u_3 &= u_3^0(x^1, x^2) + u_3^1(x^1, x^2) x^3 + \frac{1}{2} u_3^2(x^1, x^2) (x^3)^2 + \dots \end{aligned}$$

$$\sigma_{13} = \sigma_{13}^0 f(x^3)$$

$$f(x^3) = \begin{cases} \text{чётная} \\ f(\pm \frac{h}{2}) = 0 \end{cases}$$

$$f(x^3) = 1 - \left(\frac{2x^3}{h}\right)^2$$

плоская деформация $u_3 = 0$

плоское напряженное состояние $\sigma_{33} = 0$

$$u_1(x^1, x^2, x^3) = u_1^0(x^1, x^2) + u_1^1(x^1, x^2) x^3$$

$$u_2(x^1, x^2, x^3) = u_2^0(x^1, x^2) + u_2^1(x^1, x^2) x^3$$

$$u_3(x^1, x^2, x^3) = u_3^0(x^1, x^2)$$

$$\varepsilon_{ij} = \frac{1}{2} (u_{i,j} + u_{j,i}) \quad , \quad \sigma_{ij} = \lambda^* \delta_{ij} \varepsilon_{kk} + 2\mu \varepsilon_{ij}$$

$$\varepsilon_{11} = u_{1,1} = u_{1,1}^0 + u_{1,1}^1 x^3 \quad \varepsilon_{22} = u_{2,2} = u_{2,2}^0 + u_{2,2}^1 x^3$$

$$\varepsilon_{12} = \frac{1}{2} [u_{1,2}^0 + u_{2,1}^0 + x^3 (u_{1,2}^1 + u_{2,1}^1)]$$

$$\varepsilon_{13} = \frac{1}{2} [u_{1,3} + u_{3,1}] = \frac{1}{2} (u_1^1 + u_{3,1}^0) \quad , \quad \varepsilon_{33} = 0$$

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$$\sigma_{11} = \lambda^* (\varepsilon_{11} + \varepsilon_{22} + \varepsilon_{33}) + 2\mu \varepsilon_{11}$$

$$= (\lambda^* + 2\mu) (u_{1,2}^0 + x^3 u_{1,2}^1) + \lambda^* (u_{2,2}^0 + x^3 u_{2,2}^1)$$

$$\sigma_{22} = (\lambda^* + 2\mu) (u_{2,2}^0 + x^3 u_{2,2}^1) + \lambda^* (u_{1,1}^0 + x^3 u_{1,1}^1)$$

$$\sigma_{12} = \mu [u_{1,2}^0 + u_{2,1}^0 + x^3 (u_{1,2}^1 + u_{2,1}^1)]$$

$$\sigma_{13} = \mu (u_{1,3}^0 + u_{3,1}^0) f(x^3)$$

$$\sigma_{23} = \mu (u_{2,3}^0 + u_{3,2}^0) f(x^3)$$

$$\int_V \sigma_{ij}^* \delta \varepsilon_{ij} dV = \int_V p(F^i - \tilde{u}_i) \delta u_i dV + \int_S P_i \delta u_i dS$$

$$\int_V \sigma_{ij} \delta \varepsilon_{ij} dV = \int_{-L}^L \left[\int_{-\frac{h}{2}}^{\frac{h}{2}} \sigma_{ij} \delta \varepsilon_{ij} dx^3 \right] d\Omega \quad V = \Omega \times \left[-\frac{h}{2}, \frac{h}{2}\right]$$

$$\Omega \subset \mathbb{R}^2 = \mathbb{R}^2(x^1, x^2)$$

$$\sigma_{11} = \frac{E}{1-\nu^2} [u_{1,1}^0 + \nu u_{2,2}^0 + x^3 (u_{1,1}^1 + \nu u_{2,2}^1)]$$

$$\sigma_{22} = \frac{E}{1-\nu^2} [u_{2,2}^0 + \nu u_{1,1}^0 + x^3 (u_{2,2}^1 + \nu u_{1,1}^1)]$$

$$\sigma_{12} = \frac{E}{2(1+\nu)} [(u_{1,2}^0 + u_{2,1}^0) + x^3 (u_{1,2}^1 + u_{2,1}^1)]$$

$$\sigma_{13} = \frac{E}{2(1+\nu)} [u_{3,1}^0 + u_{1,3}^0]$$

$$\sigma_{23} = \frac{E}{2(1+\nu)} [u_{3,2}^0 + u_{2,3}^0]$$

$$\int_{-\frac{h}{2}}^{\frac{h}{2}} c_0 + c_1 x + c_2 x^2 dx^3 = c_0 h + c_1 (0) + c_2 \frac{h^3}{12}$$

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$\frac{3}{x} x^3 \frac{\partial}{\partial x}$
↓

$\frac{3}{x} x^3 \frac{\partial}{\partial x}$
↓

$$\int_{-\frac{h}{2}}^{\frac{h}{2}} \sigma_{11} \delta \varepsilon_{11} dx^3 = \frac{E}{1-\nu^2} \left[\underline{u_{1,1}^0 + \nu u_{2,2}^0} + \underline{x^3 (u_{1,1}^1 + u_{2,2}^1)} \right] \left(\underline{\delta u_{1,1}^0} + \underline{x^3 \delta u_{1,1}^1} \right)$$

$$= \frac{E}{1-\nu^2} (u_{1,1}^0 + \nu u_{2,2}^0) \delta u_{1,1}^0 + \frac{E h^3}{12(1-\nu^2)} (u_{1,1}^1 + \nu u_{2,2}^1) \delta u_{1,1}^1$$

интегр. по сечению
↓

$$\delta \varepsilon_{ij} = \frac{1}{2} (\delta u_{i,j} + \delta u_{j,i})$$

$$= \frac{E h}{1-\nu^2} (u_{1,1}^0 + \nu u_{2,2}^0) \delta u_1^0 + \frac{E h^3}{12(1-\nu^2)} (u_{1,1}^1 + \nu u_{2,2}^1) \delta u_1^1$$

↑
после интегр. по сечению